

Big Data Computing

Master's of Science in Computer Science

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Module 2: High-Dimensional Data Representations

Lecture 2.1: The Geometry of High-Dimensional Spaces

Philosophy of High-Dimensional Geometry

"In high dimensions, almost all of the volume of a convex body lies near its surface, and almost all pairs of points are almost orthogonal."

— Folklore of high-dimensional geometry, formalized in Blum, Hopcroft & Kannan

Our geometric intuition was built by living in three dimensions. In this lecture, we prove, step by step, exactly where and why that intuition stops being reliable.

| Lecture Roadmap

This lecture builds, from first principles, the mathematical machinery behind the **curse of dimensionality** — a phenomenon that underlies nearly every algorithm in the remainder of this course.

Specifically, we will study:

1. **Intuition Fails in High Dimensions** — framing the problem
2. **Volume Concentration in the Hypercube** — volume lives near the boundary
3. **Volume Concentration in the Hypersphere** — volume vanishes, then concentrates near the surface
4. **Concentration Near the Equator** — why random vectors are nearly orthogonal
5. **Distance Concentration** — why "near" and "far" stop meaning anything
6. **Metric Space Breakdown** — practical consequences for algorithms
7. **Synthesis** — assembling the full argument

This lecture is theoretical by design — every result is derived, not merely asserted.

| Why This Matters: From Infrastructure to Representation

Module 1 built the platforms — storage, batch computation, streaming — that big data systems run on. Module 2 now asks a different question: **once data lives on these platforms, how is it mathematically represented?**

- Text, images, and graphs are routinely represented as vectors in spaces with hundreds or thousands of dimensions.
- Many classical algorithms (nearest-neighbor search, clustering, classification) implicitly assume low-dimensional geometric intuition holds.
- Before building or trusting such algorithms at scale, we must understand precisely how geometry itself changes as dimensionality grows.

This lecture is the mathematical foundation for everything in the rest of Module 2.

§ 1 — Intuition Fails in High Dimensions

| Low-Dimensional Intuition

Most of our geometric reasoning is built from direct experience with $d = 2$ and $d = 3$:

- A circle's area is mostly "filled in," not concentrated near its edge.
- Two randomly chosen directions in 3D space are usually *not* close to perpendicular.
- "Nearest neighbor" is an obviously meaningful concept — some points really are closer than others.

Every one of these intuitions, we will show, fails dramatically once d becomes large.

| Why Intuition Is Built From 3D Experience

This is not a coincidence of mathematical taste — it is a direct consequence of biology and physics:

- Human visual and spatial reasoning evolved to navigate a three-dimensional physical world.
- We have no direct sensory experience of a 100-dimensional or 1000-dimensional space.
- As a result, intuition extrapolated naively from 2D/3D pictures can be actively **misleading** in high dimensions — not just incomplete.

| The Goal: Formalize Where Intuition Breaks

This lecture's purpose is precise: derive, with rigor, exactly **how** and **why** geometric intuition fails as dimensionality grows.

- Not "high-dimensional spaces are weird" as a vague slogan.
- Instead: specific theorems, with derivations, quantifying *how fast* volume, distance, and angle behavior diverges from low-dimensional expectation as $d \rightarrow \infty$.

| Vectors and Points in $\mathbb{R}^d$

We work throughout with the d -dimensional real vector space $\mathbb{R}^d$:

$$\mathbf{x} = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$$

- A **point** and a **vector** are used interchangeably here — a point's coordinates are simply the vector from the origin to that point.
- The Euclidean norm (length) of $\mathbf{x}$ is:

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2}$$

| Volume and Distance: The Two Central Objects of Study

This lecture's entire argument is built around two geometric quantities, tracked as d grows:

- **Volume:** how is the "mass" of a high-dimensional region (cube, sphere) distributed across that region?
- **Distance:** how spread out are the pairwise distances between random points in such a region?

We will show these two questions are deeply connected — volume concentration is, in fact, the root cause of distance concentration.

| The Unit Hypercube and Unit Hypersphere, Defined

Our two main objects of study throughout this lecture:

Unit hypercube in $\mathbb{R}^d$:

$$C_d = [0, 1]^d = \{\mathbf{x} \in \mathbb{R}^d : 0 \leq x_i \leq 1 \text{ for all } i\}$$

Unit ball (hypersphere interior) in $\mathbb{R}^d$:

$$B_d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}$$

Note that $B_d = \{\mathbf{x} \in \mathbb{R}^d : \sqrt{x_1^2 + x_2^2 + \dots + x_d^2} \leq 1\}$. Since both sides of the disequation are nonnegative, we can square them obtaining:

$$B_d = \{\mathbf{x} \in \mathbb{R}^d : x_1^2 + x_2^2 + \dots + x_d^2 \leq 1\}$$

| A First Surprising Fact, Stated Without Proof

As a preview of what this lecture will rigorously establish:

*As $d \rightarrow \infty$, the volume of the unit ball B_d approaches **zero** — even though the ball still has radius exactly 1, unchanged.*

This seems paradoxical at first glance. By the end of Section 3, we will have derived precisely why it is true.

| Why Big Data Forces High d

This is not merely a mathematical curiosity — high-dimensional spaces are the default setting in modern big data systems:

- **Text:** documents are often represented as vectors over a vocabulary of tens of thousands of words.
- **Images:** raw pixel representations, or learned embeddings, frequently have hundreds to thousands of dimensions.
- **Graphs:** node embeddings often live in spaces of dimension 64–512 or more.

The algorithms covered later in this course operate routinely in regimes where d is large enough for everything in this lecture to matter directly.

Roadmap of the Mathematical Argument Ahead

The results in this lecture build on one another in a specific logical sequence:

Volume concentrates near
the cube's boundary (§2)



Volume concentrates near
the sphere's surface (§3)



Most surface area lies
near any equator (§4)



Random vectors are nearly
orthogonal (§4)



Pairwise distances concentrate;
"nearest neighbor" loses meaning (§5)

✓ Checkpoint 1 — Framing the Problem

Let's test our basic understanding:

1. Why is our geometric intuition particularly unreliable in high dimensions?
2. What are the two central geometric quantities this lecture tracks as d grows?
3. State (without proof) the surprising fact about the volume of the unit ball as $d \rightarrow \infty$.
4. Give two examples of big data representations that routinely have high dimensionality.

Write down your thoughts before moving to Section 2.

§ 2 — Volume Concentration in the Hypercube

| Volume of the d -Dimensional Unit Cube

The unit hypercube $C_d = [0, 1]^d$ has, by definition:

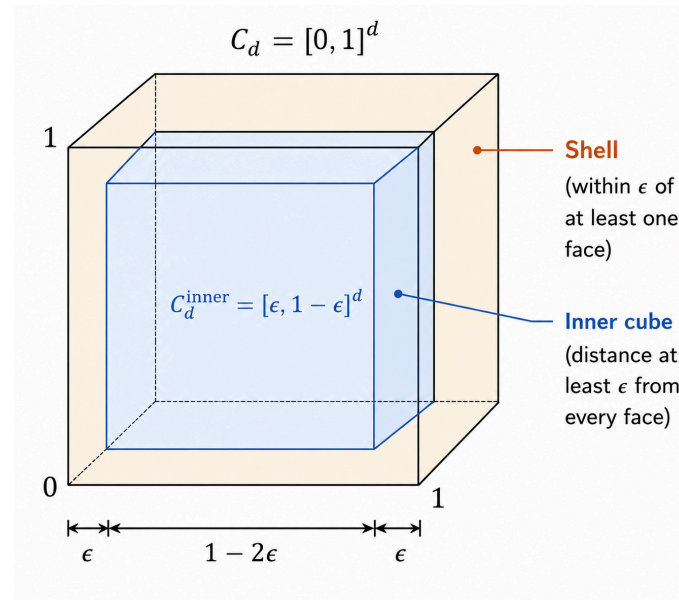
$$\text{Vol}(C_d) = 1^d = 1$$

Trivial as a standalone fact — but it gives us a fixed total volume of "1" against which we can measure how that volume is distributed throughout the cube.

The "Shell" of the Cube Near Its Boundary

Define the **inner cube** of side length $(1 - 2\epsilon)$, centered within C_d , consisting of all points at distance at least ϵ from every face:

$$C_d^{\text{inner}} = [\epsilon, 1 - \epsilon]^d$$



- The **shell** is everything in C_d but *not* in C_d^{inner} — i.e., points within ϵ of at least one face of the cube.
- *Question:* as d grows, what fraction of the cube's total volume lies in this thin shell?

| Computing the Volume of the Shrunk Inner Cube

The inner cube has side length $(1 - 2\epsilon)$ in each of d dimensions, so:

$$\text{Vol}(C_d^{\text{inner}}) = (1 - 2\epsilon)^d$$

Note that for any fixed $\epsilon \in (0, 1/2)$, the base $(1 - 2\epsilon)$ is strictly less than 1.

| Derivation: Fraction of Cube Volume Within ϵ of the Boundary

The fraction of the cube's volume contained in the boundary shell is:

$$\frac{\text{Vol}(C_d) - \text{Vol}(C_d^{\text{inner}})}{\text{Vol}(C_d)} = 1 - (1 - 2\epsilon)^d$$

Since $\text{Vol}(C_d) = 1$, this fraction equals $1 - (1 - 2\epsilon)^d$ directly.

| The Limit as $d \rightarrow \infty$

Since $0 < (1 - 2\epsilon) < 1$ for any fixed $\epsilon \in (0, 1/2)$, raising it to an increasing power d drives it to zero:

$$\lim_{d \rightarrow \infty} (1 - 2\epsilon)^d = 0$$

Therefore:

$$\lim_{d \rightarrow \infty} \left[1 - (1 - 2\epsilon)^d \right] = 1$$

*The fraction of the cube's volume in the boundary shell approaches **100%**, for any fixed $\epsilon > 0$, no matter how small.*

Interpretation: Almost All Volume Lives Near the Surface

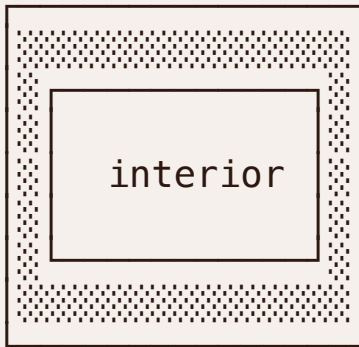
This derivation establishes our first major high-dimensional fact:

Theorem (Cube Volume Concentration). *For any fixed $\epsilon > 0$, as $d \rightarrow \infty$, the fraction of the unit cube's volume lying within ϵ of its boundary approaches 1.*

- In low dimensions ($d = 2$: a square), a "thin border strip" is genuinely a small fraction of the total area.
- In high dimensions, there is essentially **no interior** — almost the entire cube *is*, in a precise sense, its own surface.

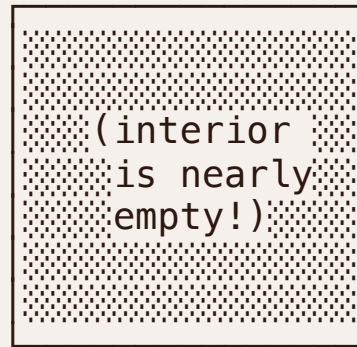
Diagram: 2D vs. High-D Cube Shell Illustration

$d = 2$ (square):



shell = small fraction

$d = \text{large}$ (conceptual):



shell $\approx$ entire volume

| Corner Concentration: Where Do Random Points Lie?

A related question: if we pick a point uniformly at random in C_d , how far is it from the cube's center, typically?

- Each coordinate x_i is drawn uniformly from $[0, 1]$, independently.
- Intuition from $d = 2, 3$ suggests a "central" region should be reasonably likely — but as just shown, this intuition fails for large d .

*The boundary-concentration result already tells us: a random point is, with high probability, close to **some** face of the cube — i.e., close to a "corner-like" region, not the center.*

| The Diagonal of the Hypercube Grows With $\sqrt{d}$

The length of the cube's main diagonal (from $\mathbf{0}$ to $(1, 1, \dots, 1)$) is:

$$\text{diag}(C_d) = \sqrt{1^2 + 1^2 + \dots + 1^2} = \sqrt{d}$$

- While each side of the cube has fixed length 1, the diagonal grows **without bound** as d increases.
- This means the cube becomes, in a precise sense, increasingly "spiky" — most of its volume is far from the center relative to its side length.

| Why This Already Hints at Distance Behavior

The growing diagonal, combined with boundary-volume concentration, foreshadows the central result of Section 5:

- If most random points sit near the cube's boundary, and the boundary itself spans a diagonal of length $\sqrt{d}$, then pairwise distances between random points should scale with $\sqrt{d}$ as well.
- This is exactly what we will derive rigorously in Section 5 — for now, simply note the structural connection between volume concentration and distance growth.

✓ Checkpoint 2 — Hypercube Volume Concentration

Let's test our understanding:

1. Derive the fraction of the unit cube's volume lying within ϵ of its boundary, for general d .
2. Why does this fraction approach 1 as $d \rightarrow \infty$, for any fixed $\epsilon > 0$?
3. What does it mean, intuitively, to say "the cube has almost no interior" in high dimensions?
4. How does the cube's diagonal length relate to d , and why is this relevant to distance behavior?

Discuss with your colleague or write down your answers.

§ 3 — Volume Concentration in the Hypersphere

| The Volume Formula for a d -Ball

The volume of a d -dimensional ball of radius r is given by the closed-form expression:

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2} + 1\right)} r^d$$

where $\Gamma(\cdot)$ is the Gamma function, the continuous generalization of the factorial ($\Gamma(n + 1) = n!$ for integer n).

This formula generalizes familiar low-dimensional cases — at $d = 2$ it gives πr^2 ; at $d = 3$ it gives $\frac{4}{3}\pi r^3$.

| Behavior of $V_d(1)$ as $d \rightarrow \infty$: Volume Vanishes

Consider the volume of the **unit** ball ($r = 1$) as a function of d :

$$V_d(1) = \frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2} + 1\right)}$$

- The numerator $\pi^{d/2}$ grows exponentially in d .
- The denominator $\Gamma(d/2 + 1)$ grows **faster than exponentially** (factorial-like growth), by Stirling's approximation ($n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$).
- The denominator therefore eventually dominates, and:

$$\lim_{d \rightarrow \infty} V_d(1) = 0$$

Confirming the Surprising Fact From Section 1

This derivation confirms the claim previewed in Section 1:

Theorem (Vanishing Ball Volume). *The volume of the d -dimensional unit ball approaches 0 as $d \rightarrow \infty$.*

- The ball's radius never changes — it remains exactly 1 throughout.
- Yet its d -dimensional volume shrinks toward zero.
- This is a direct consequence of how the normalizing Gamma function in the denominator grows.

| Worked Numeric Example: Unit Ball Volume Across Dimensions

Evaluating $V_d(1) = \pi^{d/2} / \Gamma(d/2 + 1)$ for increasing d :

d	$V_d(1)$ (approx.)	Trend
2	3.14 ($= \pi$)	Familiar circle area
3	4.19 ($= \frac{4}{3}\pi$)	Familiar sphere volume
5	5.26	Near the peak
10	2.55	Already shrinking
20	0.026	Sharp decline
50	1.7×10^{-13}	Effectively zero

Volume actually peaks around $d = 5$, then decays toward zero — a fact worth noting precisely because it shows the phenomenon is not simply monotonic from the start, even though the asymptotic limit is unambiguous.

| Why Stirling's Approximation Explains This Decline

Stirling's approximation gives, for large n : $\Gamma(n + 1) \approx \sqrt{2\pi n} (n/e)^n$. Applying this to $\Gamma(d/2 + 1)$ shows it grows roughly like $(d/2e)^{d/2}$ — a **super-exponential** function of d .

- Compare this to the numerator $\pi^{d/2}$, which grows only **exponentially** in d .
- Any super-exponential function eventually overtakes any exponential function, which is exactly why $V_d(1) \rightarrow 0$ despite initially increasing for small d .

| The Ratio of Inscribed Sphere to Circumscribing Cube Volume

Consider a unit ball B_d inscribed inside the cube $[-1, 1]^d$ (side length 2, so the ball of radius 1 fits exactly inside).

The cube's volume is 2^d . The ratio of ball volume to cube volume is:

$$\frac{V_d(1)}{2^d} = \frac{\pi^{d/2}}{2^d \Gamma\left(\frac{d}{2} + 1\right)}$$

| Derivation: This Ratio $\rightarrow 0$ as $d \rightarrow \infty$

Since $V_d(1) \rightarrow 0$ (just shown) while 2^d grows without bound, the ratio:

$$\frac{V_d(1)}{2^d} \longrightarrow 0 \quad \text{as } d \rightarrow \infty$$

The denominator's super-exponential growth (from the Gamma function) combined with the additional exponential factor 2^d drives this ratio to zero extremely quickly.

| Interpretation: Spheres Become Negligible Inside Cubes

This result has a striking geometric interpretation:

- In $d = 2$: a circle inscribed in a square fills roughly 78.5% of the square's area ($\pi/4$).
- In $d = 3$: an inscribed sphere fills roughly 52.4% of the cube's volume ($\pi/6$).
- As d grows large, the inscribed ball occupies an **almost negligible fraction** of the cube's volume — nearly all of the cube's volume lies in its "corners," outside the inscribed sphere entirely.

| Volume of a Thin Spherical Shell

Now, paralleling Section 2's cube-shell analysis, consider a thin shell near the **surface** of the ball itself: the region between radius $(1 - \epsilon)$ and radius 1.

Since $V_d(r) = V_d(1) \cdot r^d$ (volume scales with r^d), the inner ball of radius $(1 - \epsilon)$ has volume:

$$V_d(1 - \epsilon) = V_d(1) \cdot (1 - \epsilon)^d$$

| Derivation: Fraction of Ball's Volume Within ϵ of the Surface

The fraction of the ball's volume lying in the shell (within ϵ of the surface) is:

$$\frac{V_d(1) - V_d(1 - \epsilon)}{V_d(1)} = 1 - (1 - \epsilon)^d$$

This is structurally identical to the cube-shell derivation in Section 2 — only the base of the exponent differs slightly ($1 - \epsilon$ here vs. $1 - 2\epsilon$ for the cube).

| The Limit: Nearly All Volume Concentrates Near the Surface

By the same reasoning as Section 2, for any fixed $\epsilon \in (0, 1)$:

$$\lim_{d \rightarrow \infty} (1 - \epsilon)^d = 0 \quad \implies \quad \lim_{d \rightarrow \infty} \left[1 - (1 - \epsilon)^d \right] = 1$$

Theorem (Sphere Surface Concentration). *For any fixed $\epsilon > 0$, as $d \rightarrow \infty$, the fraction of the ball's volume lying within ϵ of its surface approaches 1.*

Even though the ball's total volume itself is vanishing (previous result), what little volume remains is concentrated almost entirely in a thin shell near the surface.

Comparing Cube-Shell and Sphere-Shell Concentration: The Same Underlying Phenomenon

Both Section 2 (cube) and this section (sphere) arrive at structurally identical results:

Object	Fraction in boundary shell	Limit as $d \rightarrow \infty$
Hypercube	$1 - (1 - 2\epsilon)^d$	$\rightarrow 1$
Hypersphere	$1 - (1 - \epsilon)^d$	$\rightarrow 1$

Both derivations rely on the same core mathematical fact: any fixed base strictly between 0 and 1, raised to a growing power d , vanishes. This single fact is the engine behind nearly every result in this lecture.

✓ Checkpoint 3 — Hypersphere Volume Concentration

Let's test our understanding:

1. Why does the volume of the unit ball $V_d(1)$ approach 0 as $d \rightarrow \infty$, despite its radius staying fixed at 1?
2. Derive the fraction of the ball's volume lying within ϵ of its surface, for general d .
3. What is the geometric interpretation of the ratio of inscribed-sphere to circumscribing-cube volume vanishing?
4. What single mathematical fact underlies both the cube-shell and sphere-shell concentration results?

Review your answers before moving on.

§ 4 — Concentration Near the Equator

| A New Question: Where on the Sphere's Surface Do Points Concentrate?

Section 3 established that points on a random ball concentrate near its **surface** (the sphere itself). This section asks a sharper question:

- Given that a random point lies on the surface of the unit sphere, **where on that surface** is it likely to be?
- Specifically: fix any direction (a "pole"), defining an "equator" perpendicular to it. How close to this equator does a random point on the sphere tend to lie?

| Defining a "Band" Near the Equator

Fix a unit vector $\mathbf{e}$ (the "north pole" direction). For a point $\mathbf{x}$ on the unit sphere, define its angular position via:

$$x_1 = \mathbf{x} \cdot \mathbf{e}$$

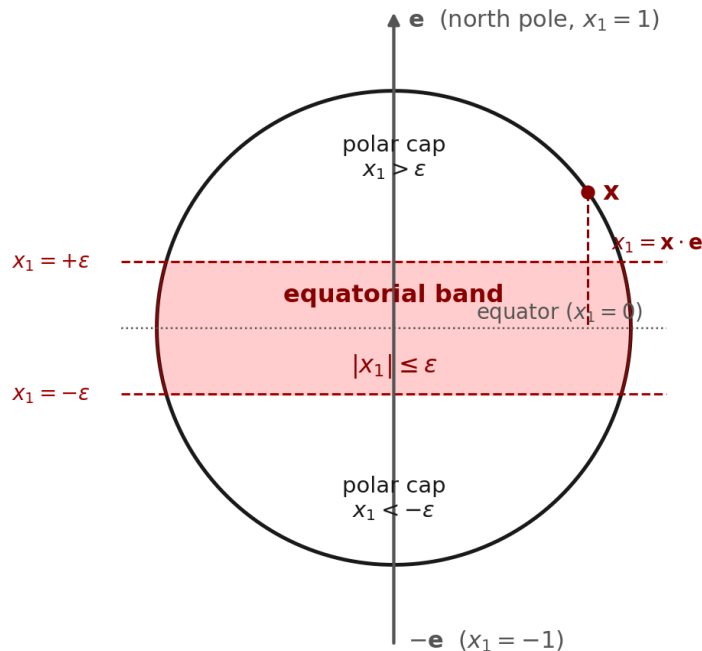
(its coordinate along the $\mathbf{e}$ direction). The **equatorial band** of width ϵ is the set of points with $|x_1| \leq \epsilon$.

Question: as $d \rightarrow \infty$, what fraction of the sphere's surface area lies within this band?

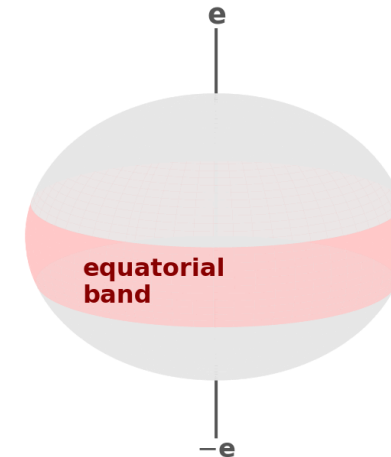
Picturing the Equatorial Band

The Equatorial Band $\{ \mathbf{x} : |\mathbf{x} \cdot \mathbf{e}| \leq \epsilon \}$

2D cross-section through the pole axis



Full sphere in $\mathbb{R}^d$ (shown as $d = 3$)



- **Left:** a 2D cross-section through the pole axis $\mathbf{e}$. The band is the horizontal strip $|x_1| \leq \epsilon$; everything above/below it is a **polar cap**.
- **Right:** the same band on the full sphere (shown at $d = 3$ for intuition — the argument itself holds for any d).

| Sketching Why the Sphere Concentrates Near the Equator

Consider the sphere $S_{d-1} = \{x \in \mathbb{R}^d : x_1^2 + x_2^2 + \cdots + x_d^2 = 1\}$. This is just the **surface** of the corresponding d -dimensional ball B_d .

The sphere S_{d-1} has **intrinsic dimension** $(d - 1)$: one constraint in d coordinates.

Now fix the first coordinate: $x_1 = t$.

Substituting into the sphere equation gives $x_2^2 + \cdots + x_d^2 = 1 - t^2$.

Thus the slice is a sphere in the remaining $(d - 2)$ coordinates: S_{d-2} with radius $r = \sqrt{1 - t^2}$.

So fixing one coordinate further reduces the intrinsic dimension by one:

$$\boxed{(d - 1) \longrightarrow (d - 2)}.$$

Example: $(d = 3)$.

The sphere $S_2 \subset \mathbb{R}^3$ is a 2-dimensional surface. Fixing $x_1 = t$ gives $x_2^2 + x_3^2 = 1 - t^2$, which is a **circle** S_1 of radius $\sqrt{1 - t^2}$.

| Why Does the Slice Get Smaller Away From the Equator?

A $(d - 2)$ -dimensional sphere of radius r has surface area proportional to r^{d-2} .

Since $r = \sqrt{1 - t^2}$, the area of the slice at $x_1 = t$ is therefore proportional to

$$\left(\sqrt{1 - t^2}\right)^{d-2} = (1 - t^2)^{(d-2)/2}.$$

At the equator, $t = 0$:

$$(1 - t^2)^{(d-2)/2} = 1.$$

Away from the equator, $|t| > 0$, we have

$$0 < 1 - t^2 < 1.$$

Therefore, as d increases,

$$(1 - t^2)^{(d-2)/2} \longrightarrow 0.$$

So slices farther from the equator become **exponentially smaller in their surface area**.

High-dimensional geometry strongly favors the equator:

there is much more surface area available near $x_1 = 0$ than near the poles.

| From Geometric Intuition to Concentration

The previous slide explains **why** the equator should dominate:

slightly smaller radius $\implies$ dramatically smaller area in high dimensions.

The precise concentration result is:

$$\Pr [|x_1| > \epsilon] \leq e^{-d\epsilon^2/2}.$$

Therefore, for any fixed $\epsilon > 0$,

$$e^{-d\epsilon^2/2} \longrightarrow 0 \quad \text{as } d \rightarrow \infty.$$

Hence,

$$\boxed{\Pr [|x_1| \leq \epsilon] \longrightarrow 1.}$$

In words:

As the dimension grows, almost all of the sphere's surface lies within any fixed-width equatorial band.

The band does **not** become wider. Rather, the surface area becomes increasingly concentrated inside it.

| Why This Generalizes to *Any* Great Hypersphere Through the Center

A crucial subtlety: this result holds for **any** choice of the "pole" direction $\mathbf{e}$, not just one special axis:

- By the rotational symmetry of the sphere, the choice of which direction counts as "north" is arbitrary.
- This means: for **any** fixed direction we pick in advance, a random point on the sphere is overwhelmingly likely to lie near the equator perpendicular to that direction.

| Consequence: Two Random Points Are Almost Orthogonal

This generalization leads directly to one of the most counterintuitive results in this lecture. Suppose $\mathbf{x}$ and $\mathbf{y}$ are two **independent** random points on the unit sphere in $\mathbb{R}^d$.

- Fix $\mathbf{y}$, and treat it as the "pole" direction $\mathbf{e}$.
- By the equatorial concentration result, $\mathbf{x} \cdot \mathbf{y}$ is, with high probability, close to 0.
- But $\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos(\theta)$, where θ is the angle between $\mathbf{x}$ and $\mathbf{y}$
- Since $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$, $\mathbf{x} \cdot \mathbf{y} = \cos(\theta)$, so $\cos(\theta) \approx 0$ means $\theta \approx 90^\circ$.

Formal Statement: Random Unit Vectors Are Nearly Orthogonal in High d

Theorem (Near-Orthogonality). For two independent, uniformly random unit vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$:

$$\Pr [|\mathbf{x} \cdot \mathbf{y}| > \epsilon] \leq e^{-d\epsilon^2/2}$$

As $d \rightarrow \infty$, this probability vanishes — meaning $\mathbf{x}$ and $\mathbf{y}$ are, with overwhelming probability, **almost exactly orthogonal**.

In low dimensions ($d = 2, 3$), two random directions are typically nowhere near perpendicular. In high dimensions, near-perpendicularity becomes the overwhelmingly typical case.

| Connecting Back to Volume Concentration as the Same Root Cause

This near-orthogonality result is not a separate phenomenon from Sections 2-3 — it follows from the **same** underlying mechanism:

- Section 2/3: a fixed-base exponential term $(1 - \epsilon)^d \rightarrow 0$ drives volume to concentrate near boundaries/surfaces.
- Section 4: an analogous exponential decay $e^{-d\epsilon^2/2} \rightarrow 0$ drives surface area to concentrate near any fixed equator.

In each case, increasing dimensionality d acts as the driving force behind an exponential concentration effect — this is the unifying mathematical theme of this entire lecture.

✓ Checkpoint 4 — Equatorial Concentration & Orthogonality

Let's test our understanding:

1. What does it mean for surface area to "concentrate near the equator," and why does this hold for *any* chosen pole direction?
2. Why does equatorial concentration imply that two random unit vectors are nearly orthogonal in high dimensions?
3. State the near-orthogonality theorem precisely, including how the probability bound depends on d .
4. What single underlying mathematical mechanism connects this section's result to Sections 2 and 3?

Consolidate your answers before proceeding.

§ 5 — Distance Concentration & the Curse of Dimensionality

| Setup: Random Points in $[0, 1]^d$

Consider two points $\mathbf{x}, \mathbf{y}$ drawn independently and uniformly at random from the unit cube $[0, 1]^d$. We study the behavior of the Euclidean distance between them as d grows.

$$\mathbf{x} = (x_1, \dots, x_d), \quad \mathbf{y} = (y_1, \dots, y_d), \quad x_i, y_i \sim \text{Uniform}(0, 1) \text{ i.i.d.}$$

| Euclidean Distance Formula Recap

The squared Euclidean distance between $\mathbf{x}$ and $\mathbf{y}$ is:

$$\|\mathbf{x} - \mathbf{y}\|_2^2 = \sum_{i=1}^d (x_i - y_i)^2$$

This is simply a sum of d independent, identically distributed terms — one per coordinate — which makes it directly amenable to standard probabilistic tools (expectation and variance).

| Expected Squared Distance Between Two Random Points

By linearity of expectation:

$$\mathbb{E} [\|\mathbf{x} - \mathbf{y}\|_2^2] = \sum_{i=1}^d \mathbb{E} [(x_i - y_i)^2] = d \cdot \mathbb{E} [(x_1 - y_1)^2]$$

For x_1, y_1 independent and $\text{Uniform}(0, 1)$, a standard calculation gives $\mathbb{E}[(x_1 - y_1)^2] = \frac{1}{6}$.

Expected Squared Distance Between Two Random Points

Let x_1, y_1 be two random variables, such that $x_1, y_1 \sim \text{Uniform}(0, 1)$. The expected squared difference is

$$\mathbb{E}[(x_1 - y_1)^2].$$

Notice that $(x_1 - y_1)^2 = x_1^2 - 2x_1y_1 + y_1^2$. Therefore, $\mathbb{E}[(x_1 - y_1)^2] = \mathbb{E}[x_1^2] - 2\mathbb{E}[x_1y_1] + \mathbb{E}[y_1^2]$.

For any random variable $Z \sim \text{Uniform}(0, 1)$:

$$\mathbb{E}[Z] = \int_0^1 z \, dz = \frac{z^2}{2} \Big|_0^1 = \frac{1}{2}, \quad \mathbb{E}[Z^2] = \int_0^1 z^2 \, dz = \frac{z^3}{3} \Big|_0^1 = \frac{1}{3}.$$

So, $\mathbb{E}[x_1^2] = \mathbb{E}[y_1^2] = \frac{1}{3}$. Furthermore, independence gives

$$\mathbb{E}[x_1y_1] = \mathbb{E}[x_1]\mathbb{E}[y_1] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Therefore,

$$\mathbb{E}[(x_1 - y_1)^2] = \frac{1}{3} - 2 \left(\frac{1}{4} \right) + \frac{1}{3} = \boxed{\frac{1}{6}}.$$

Two independent points chosen uniformly in $[0, 1]$ are, on average, separated by a squared distance of $\frac{1}{6}$.

| Derivation: $E[\text{dist}^2]$ Grows Linearly With d

Combining the previous slide's results:

$$E [\|\mathbf{x} - \mathbf{y}\|_2^2] = \frac{d}{6}$$

Result. The expected squared Euclidean distance between two random points in $[0, 1]^d$ grows **linearly** in d .

This matches the diagonal-length intuition from Section 2: as d grows, typical distances between points grow roughly proportionally to $\sqrt{d}$.

| Variance of the Distance, and Its Relative Behavior

Since $\|\mathbf{x} - \mathbf{y}\|_2^2$ is a sum of d i.i.d. terms, its **variance** also grows linearly in d (variance of a sum of independent terms is the sum of variances):

$$\text{Var} [\|\mathbf{x} - \mathbf{y}\|_2^2] = d \cdot \text{Var} [(x_1 - y_1)^2] = d \cdot c$$

for some constant $c > 0$ (a fixed property of the Uniform(0, 1) distribution, independent of d).

| Derivation: The *Relative Spread* of Distances Shrinks as d Grows

The standard deviation of $\|\mathbf{x} - \mathbf{y}\|_2^2$ grows as $\sqrt{d \cdot c} = O(\sqrt{d})$, while its **mean** grows as $O(d)$. So the **coefficient of variation** (relative spread) behaves as:

$$\frac{\text{Std. Dev.}}{\text{Mean}} = \frac{O(\sqrt{d})}{O(d)} = O\left(\frac{1}{\sqrt{d}}\right) \longrightarrow 0 \text{ as } d \rightarrow \infty$$

The absolute spread of distances grows — but it grows much more slowly than the mean distance itself. Relative to the typical distance, the spread shrinks toward zero.

Formal Distance Concentration Result

Theorem (Distance Concentration). For points drawn independently and uniformly from $[0, 1]^d$, the squared Euclidean distance $\|\mathbf{x} - \mathbf{y}\|_2^2$ concentrates around its mean $d/6$: the ratio of standard deviation to mean vanishes as $d \rightarrow \infty$.

By standard concentration inequalities (e.g., Chebyshev's inequality), this can be tightened into explicit probabilistic bounds.

| Making This Precise With Chebyshev's Inequality

Recall Chebyshev's inequality: for a random variable Z with mean μ and variance σ^2 ,

$$\Pr [|Z - \mu| \geq k\sigma] \leq \frac{1}{k^2}$$

Applying this with $Z = \|\mathbf{x} - \mathbf{y}\|_2^2$, $\mu = d/6$, and $\sigma = O(\sqrt{d})$:

$$\Pr \left[\left| \|\mathbf{x} - \mathbf{y}\|_2^2 - d/6 \right| \geq k\sqrt{d} \right] \leq \frac{1}{k^2}$$

This gives an explicit, quantitative version of "distances concentrate around their mean" — the deviation from the mean is bounded in terms of $\sqrt{d}$, which is small relative to the mean's own size of $d/6$.

| Interpreting the Chebyshev Bound at Concrete Dimensions

Setting $k = 10$ (so the bound guarantees at least 99% probability mass within 10σ of the mean), and recalling $\sigma = O(\sqrt{d})$:

d	Mean $\mu = d/6$	Bound width 10σ	Relative width $10\sigma/\mu$
10	1.67	≈ 12.9	Large — bound is not yet tight
1,000	166.7	≈ 129	Large in absolute terms...
1,000	166.7	—	...but $\sigma/\mu = O(1/\sqrt{d}) \rightarrow 0$ regardless

The qualitative conclusion — relative spread shrinking as $1/\sqrt{d}$ — holds regardless of the specific constant k chosen; Chebyshev simply makes the bound rigorous and explicit.

| Interpretation: Max Distance / Min Distance $\rightarrow 1$

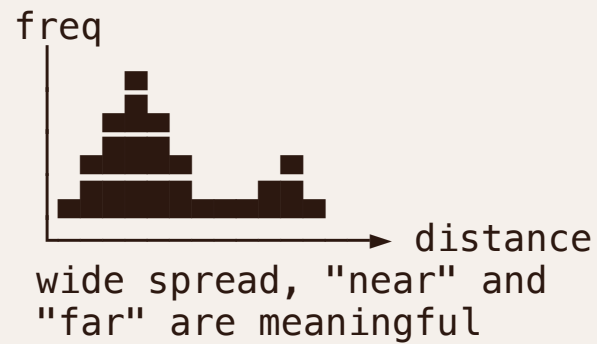
A striking practical consequence of distance concentration:

$$\frac{\max_{i,j} \|\mathbf{x}_i - \mathbf{x}_j\|_2}{\min_{i,j} \|\mathbf{x}_i - \mathbf{x}_j\|_2} \longrightarrow 1 \quad \text{as } d \rightarrow \infty$$

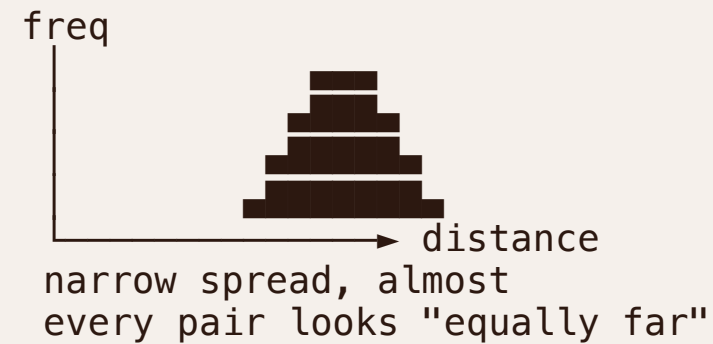
for any fixed finite set of random points. *In high dimensions, the farthest pair of points and the nearest pair of points become, relatively speaking, almost equally far apart.*

Diagram: Histogram of Pairwise Distances, Low-D vs. High-D

Low d: distances vary widely



High d: distances cluster tightly



| Why This Breaks the Notion of "Nearest Neighbor"

This concentration result has a direct, practical consequence for algorithms relying on distance:

- A "nearest neighbor" search assumes some points are *meaningfully* closer to a query than others.
- If all pairwise distances are nearly equal (relative to their scale), the **nearest** neighbor is barely closer than a **typical** or even a relatively **far** neighbor.
- The entire notion of "closeness" as a discriminative signal degrades as dimensionality grows.

| Worked Numeric Example: Small d vs. Large d

Illustrative (not exact simulation) values showing the qualitative trend, for points drawn uniformly in $[0, 1]^d$:

d	Mean squared distance ($d/6$)	Relative spread (std/mean)	Practical effect
2	0.33	High	Distances vary noticeably; "near" is meaningful
10	1.67	Moderate	Some discrimination remains
100	16.7	Low	Distances cluster tightly around the mean
1000	166.7	Very low	Nearest and farthest neighbor barely differ

| The Curse of Dimensionality, Named and Defined Precisely

We can now state, precisely, what the term **curse of dimensionality** refers to in this context:

Curse of Dimensionality. *As the dimensionality d of a feature space grows, the relative discriminative power of distance metrics — specifically, the ability to distinguish "near" points from "far" points — degrades toward zero, as a direct consequence of volume and distance concentration.*

This is the precise, derivable phenomenon underlying the informal phrase used throughout the data mining and machine learning literature.

✓ Checkpoint 5 — Distance Concentration

Let's test our understanding:

1. Derive the expected squared Euclidean distance between two random points in $[0, 1]^d$, and explain why it grows linearly in d .
2. Why does the *relative* spread of pairwise distances shrink as d grows, even though the *absolute* spread grows?
3. What does it mean for the ratio of maximum to minimum pairwise distance to approach 1?
4. State the curse of dimensionality precisely, in terms of distance discrimination.

Make sure you understand these derivations before wrapping up this section.

§ 6 — Metric Space Breakdown & Practical Consequences

| What "Breakdown" Means Formally

We say a distance metric **breaks down** in high dimensions when it loses its **discriminative power** — its ability to meaningfully separate "similar" from "dissimilar" points:

- The metric itself remains mathematically well-defined (it still satisfies the triangle inequality, non-negativity, etc.).
- What breaks down is its **usefulness** as a signal: the relative gap between near and far distances vanishes, as shown in Section 5.

| Euclidean (L_2) Distance Specifically: Why It Degrades

The Euclidean (L_2) distance is particularly susceptible to this breakdown because it aggregates information across **all** d coordinates via a sum of squares:

$$\|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$$

- Each coordinate contributes independent "noise" to the total.
- As d grows, this accumulated noise — by the Law of Large Numbers-style averaging behavior derived in Section 5 — dominates over any genuine, meaningful signal in a small subset of relevant coordinates.

| Do Other Metrics Fare Better? A Brief Comparative Note

The same fundamental concentration phenomenon affects other common distance metrics, though not always identically:

- L_1 **(Manhattan) distance:** $\sum_i |x_i - y_i|$ — exhibits the same qualitative concentration behavior as L_2 , though some empirical studies suggest a somewhat slower relative degradation.
- L_∞ **(Chebyshev) distance:** $\max_i |x_i - y_i|$ — concentrates differently, since it depends on only the single largest coordinate-wise difference.

This is an active and nuanced area of research; the qualitative lesson — that high dimensionality degrades distance-based discrimination broadly — holds across standard metrics, even if the precise rate varies.

| Consequences for Nearest-Neighbor Search

Recall the data mining context from later in this course: nearest-neighbor and similarity search are foundational operations.

- If nearest and farthest neighbors become nearly indistinguishable (Section 5), naive nearest-neighbor search in raw high-dimensional space provides little real discriminative value.
- This motivates two major directions covered later in the course: **dimensionality reduction** (Lecture 2.3) to reduce d while preserving meaningful structure, and **Locality-Sensitive Hashing** (Lecture 3.5) to search efficiently despite high dimensionality.

| Consequences for Clustering Algorithms

Many clustering algorithms (e.g., k -means) fundamentally rely on Euclidean distance to define cluster membership:

- k -means assigns each point to its nearest centroid — but if "nearest" barely differs from "average distance" (Section 5), cluster assignments become unstable and less meaningful.
- In high-dimensional feature spaces, naive Euclidean clustering can produce clusters that do not reflect any real underlying structure in the data.

| Consequences for Classification With Euclidean-Based Methods

Classification methods relying directly on Euclidean distance (e.g., k -nearest-neighbor classifiers) face the same fundamental issue:

- A test point's " k nearest" labeled training points may be only marginally closer than a much larger set of "moderately near" points.
- Classification accuracy can degrade as raw feature dimensionality grows, even when more features should, in principle, carry more information — this is a direct manifestation of the curse of dimensionality.

| Why Naive Distance-Based ML Breaks at Scale in High-D Feature Spaces

Bringing together this section's consequences:

- It is not that high-dimensional data is inherently unusable — it is that **naive, unmodified** application of low-dimensional-intuition algorithms (raw Euclidean nearest-neighbor, raw Euclidean clustering) systematically underperforms as d grows.
- Big data systems, by their nature, frequently produce exactly this kind of high-dimensional representation (Section 1) — making this issue a first-order concern, not a corner case.

| This Motivates Dimensionality Reduction

The natural response to this lecture's results: if raw high-dimensional distances lose their discriminative power, can we **reduce** dimensionality while **preserving** the geometric structure (distances) that actually matters?

- This is precisely the subject of the next lecture: the **Johnson-Lindenstrauss Lemma**, which shows that random projections can reduce dimensionality dramatically while approximately preserving pairwise distances.

✓ Checkpoint 6 — Metric Breakdown & Consequences

Let's test our understanding:

1. What does it mean, precisely, for a distance metric to "break down" in high dimensions?
2. Why is Euclidean (L_2) distance particularly susceptible to this phenomenon?
3. Give two concrete examples of algorithms whose performance is degraded by the curse of dimensionality, and explain why.
4. What general strategy does this lecture's results motivate for addressing the curse of dimensionality?

Consolidate your answers before the final synthesis.

§ 7 — Synthesis & Wrap-up

The Chain of Results Recap Diagram

§2: Cube volume concentrates near the boundary
 $(1-2\varepsilon)^d \rightarrow 0$

§3: Sphere volume concentrates near the surface (and vanishes in absolute terms): $(1-\varepsilon)^d \rightarrow 0$

§4: Surface area concentrates near any equator
 $e^{-d\varepsilon^2/2} \rightarrow 0$

§4: Random unit vectors are nearly orthogonal

§5: Pairwise distances concentrate around their mean
(relative spread $\rightarrow 0$ as $d \rightarrow \infty$)

§6: Distance-based algorithms lose discriminative power – the curse of dimensionality

| Summary of Key Theorems and Derivations

This lecture established, with full derivations, a connected chain of results:

1. **Cube/sphere volume concentration:** almost all volume lies near the boundary/surface as $d \rightarrow \infty$.
2. **Vanishing ball volume:** the unit ball's volume itself shrinks to zero, even at fixed radius 1.
3. **Equatorial concentration & near-orthogonality:** random unit vectors become almost perpendicular.
4. **Distance concentration:** pairwise distances cluster tightly around their mean, relative to their scale.
5. **The curse of dimensionality:** distance-based algorithms lose discriminative power as a direct, provable consequence of 1-4.

| Key Terms Glossary

Term	Meaning
Volume concentration	Almost all volume of a cube/ball lies near its boundary/surface in high d
Equatorial concentration	Surface area of a high- d sphere concentrates near any fixed equator
Near-orthogonality	Random unit vectors in high d are almost exactly perpendicular
Distance concentration	Pairwise distances cluster tightly around their mean as $d \rightarrow \infty$
Curse of dimensionality	Loss of distance-based discriminative power as d grows

| Connecting to Module 1: Callback

This lecture's results are not abstract mathematics disconnected from the rest of the course — they describe the geometry of exactly the kind of data Module 1's infrastructure was built to store and process:

- Vectors with hundreds or thousands of dimensions (text embeddings, image features, graph embeddings — Lectures 2.4-2.6) are routinely stored as blocks across HDFS/GFS (Lecture 1.1) and processed via Spark or MapReduce (Lectures 1.2-1.3).
- Any algorithm operating on this data — clustering, classification, similarity search — must contend with exactly the curse of dimensionality derived in this lecture.

| Preview: Lecture 2.2 — Dimensionality Reduction & Random Projections

Having established precisely *why* high-dimensional distances lose their meaning, the next lecture asks: can we do something about it?

- **The Johnson-Lindenstrauss Lemma:** a random projection into a much lower-dimensional space can *approximately preserve* pairwise distances, with high probability.
- **Random projection matrices:** the mechanics of how such a projection is constructed.
- **Error bounds:** precisely how much distortion is introduced, and how it depends on the target dimension.

This lecture proved the disease. The next lecture builds the cure. See you in Lecture 2.2!

| Lecture 2.1 Complete

Recommended Reading

- Blum, A., Hopcroft, J., & Kannan, R. (2020). *Foundations of Data Science*, Ch. 2.
- Course reference materials on high-dimensional geometry

Contact & Q&A

- **Course:** Big Data Computing
- **Office Hours:** Drop me a message, and we'll arrange a time to meet, either in person or online! 😊

Thank you for your attention!